Predicting Student Academic Performance Using Big Data Analytics

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Abstract*—Predicting student academic performance has emerged as a significant research area within educational analytics, driven by the increasing availability of large-scale student data. This study aims to develop a robust Big Data model to forecast students' academic outcomes by leveraging demographic, behavioral, and academic features. Utilizing the Student Performance Data Set from the UCI Machine Learning Repository, this research applies machine learning algorithms—including Random Forest, Logistic Regression, and Gradient Boosting—within an Apache Spark framework to handle data scalability. The models are evaluated using performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC, with cross-validation employed to ensure reliability. The experimental results indicate that Gradient Boosting achieved the highest predictive performance, identifying key at-risk indicators such as previous failures and study time. The findings are expected to assist educational institutions in the early identification of at-risk students, enabling timely, data-informed interventions and personalized support mechanisms. This study explore the potential of Big Data analytics to enhance educational decision-making, improve student retention, and foster academic success.

***Keywords*—Big Data, Academic Performance, Machine Learning, Prediction Model, Education Analytics, Apache Spark**

I. INTRODUCTION

The pursuit of academic excellence and student success is a primary objective for higher education institutions worldwide. However, student attrition and academic underperformance remain persistent challenges, leading to negative outcomes for both individuals and institutions [1]. The ability to proactively identify students who are at risk of failing or dropping out is crucial for implementing effective support systems and improving retention rates.

The advent of Big Data technologies has unlocked new possibilities for analyzing complex educational datasets [2]. By applying machine learning (ML) algorithms to features encompassing student demographics, behavioral patterns, and academic history, it is possible to build predictive models that can flag at-risk students early in their academic journey [3]. This shift from reactive to proactive intervention represents a significant advancement in educational strategy.

The primary objective of this research is to design, implement, and evaluate a scalable Big Data model for predicting student academic performance. The main contributions of this work are:

1. The application of multiple ML algorithms on a curated educational dataset, preprocessed and scaled using Apache Spark.
2. A comparative analysis of model performance to identify the most effective algorithm for this prediction task.
3. The identification of the most influential features correlated with student success or failure.
4. The provision of a data-driven framework that institutions can adapt for early warning systems.

II. RELATED WORK

The application of data mining and machine learning in education, often termed Educational Data Mining (EDM) or Learning Analytics, has been extensively explored. Early work by [4] demonstrated the use of decision trees and neural networks to predict student success, highlighting factors like prior academic achievement and engagement metrics.

More recent studies have leveraged larger and more diverse datasets. For instance, [5] used a combination of institutional data and online learning activities to predict student performance with high accuracy using a Random Forest classifier. Similarly, [6] applied support vector machines (SVM) to demographic and grades data, emphasizing the importance of feature selection.

The transition to Big Data paradigms in education is a growing trend. Research by [7] utilized Hadoop and MapReduce to analyze massive logs of student interactions in Virtual Learning Environments (VLEs). Apache Spark has gained popularity for its in-memory processing capabilities, which significantly speed up iterative ML tasks [8]. Studies like [9] have begun to implement Spark MLlib for large-scale student modeling.

While these studies provide a strong foundation, many are either limited in scale or do not provide a direct comparison of multiple modern algorithms within a distributed computing framework. This study seeks to fill this gap by systematically evaluating Random Forest, Logistic Regression, and Gradient Boosting using Spark, thereby contributing a scalable and practical model for academic performance prediction.

III. RESEARCH METHODOLOGY

This section outlines the dataset, preprocessing steps, machine learning models, and evaluation framework used in this study.

A. Dataset

The primary dataset used is the Student Performance Data Set from the UCI Machine Learning Repository [10]. This dataset contains student achievement data from secondary education, featuring attributes such as:
- Demographic: Sex, Age, Address type, Family size, Parental cohabitation status, Mother's/Father's education, Parental job.
- Social & Behavioral: Going out with friends, Weekly alcohol consumption, Health status.
- Academic: Past failures, Extra educational support, Absences, First period grade (G1), Second period grade (G2).

The target variable is the final grade (G3), which was binarized into a binary classification problem: "Pass" (G3 ≥ 10) and "Fail" (G3 < 10), a common threshold in the literature [4], [10].

B. Data Preprocessing

To prepare the data for modeling and simulate Big Data conditions, the dataset was artificially augmented and processed using Apache Spark's Data Frame API for scalability.

1. Handling Missing Values: Any missing values were imputed using the median for numerical features and the mode for categorical features.
2. Feature Engineering: The final grade (G3) was transformed into the binary target variable.
3. Scaling and Encoding: Numerical features (e.g., absences, study time) were standardized using Standard Scaler. Categorical features (e.g., parental education, job) were converted into numerical format using String Indexer and One Hot Encoder.
4. Data Splitting: The preprocessed data was split into training (70%) and testing (30%) sets.

C. Machine Learning Models

Three supervised learning algorithms were selected for their proven efficacy in classification tasks:
- Logistic Regression (LR): A linear model serving as a baseline for performance comparison.
- Random Forest (RF): An ensemble method that builds multiple decision trees and aggregates their results to reduce overfitting.
- Gradient Boosting (GB): Another powerful ensemble technique that builds trees sequentially, with each new tree correcting the errors of the previous one.

All models were implemented using the MLlib library within Apache Spark.

D. Evaluation Metrics

To comprehensively evaluate model performance, the following metrics were used on the held-out test set:
- Accuracy
- Precision
- Recall
- F1-Score
- Area Under the Receiver Operating Characteristic Curve (ROC-AUC)

A 5-fold cross-validation was performed on the training set to tune hyperparameters and ensure the models' generalizability.

IV. EXPERIMENTAL RESULTS AND EVALUATION

The performance of the three machine learning models is summarized in Table I.

TABLE I. TABLE I. MODEL PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1-Score	Roc-Auc
Logistic Regression-	0.78	0.75	0.72	0.73	0.81
Random Forest	0.84	0.82	0.80	0.81	0.89
Gradient Boosting	0.87	0.85	0.83	0.84	0.92

As shown, Gradient Boosting achieved the highest performance across all metrics, followed by Random Forest. The ROC-AUC scores, which measure the model's ability to distinguish between classes, were particularly strong for the ensemble methods.

V. DISCUSSION

The results demonstrate that machine learning models, particularly advanced ensemble methods like Gradient Boosting, can effectively predict student academic performance. The high feature of importance of "past failures" and "first period grade" suggests that interventions should be targeted very early in a course or even in preceding semesters.

The superior performance of Gradient Boosting can be attributed to its sequential learning process, which effectively captures complex, non-linear relationships in the data that simpler models like Logistic Regression might miss. While Random Forest also performed well, its bagging approach was slightly less effective than boosting for this specific dataset.

A key limitation of this study is the use of a single, relatively small public dataset. Although Spark was used to demonstrate scalability, the underlying data volume was limited. Future work should validate these models on larger, multi-institutional datasets containing more diverse features, such as VLE interaction logs and socioeconomic indicators. Furthermore, exploring deep learning models or addressing potential model bias to ensure fairness are critical next steps.

VI. CONCLUSION AND FUTURE WORK

This study successfully developed a Big Data analytics framework for predicting student academic performance. By applying and comparing multiple machine learning algorithms within the Apache Spark ecosystem, it was determined that the Gradient Boosting model is most effective for this task, with key predictors being historical grades and past failures. This work provides a foundational model that educational institutions can adapt to build early warning systems, thereby facilitating timely support for at-risk students and improving overall academic outcomes.

For future work, we plan to:
1. Source and integrate larger, multi-modal datasets from partner institutions.
2. Explore deep learning architectures, such as Recurrent Neural Networks (RNNs), for modeling temporal sequences of student activity.
3. Investigate fairness and bias in the models to ensure equitable predictions across different demographic groups.
4. Develop a real-time dashboard interface for educators to interact with the model's predictions.

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